

Investment Strategy Challenge 2026 · Proposal for NKE Private Wealth (Chak family)

Asian Digital Transformation Autocallable Note

6.8% a year from Asia's semiconductor build-out, with 100% of capital returned at maturity

USD 100mn · 5-year note issued by Natixis · 6.8% p.a. memory coupon · 100% capital protected · Trade date 17 Sep 2026

Agenda

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Client & market

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Executive summary

One note that turns the family's operating expertise into protected returns

6.8%

Memory coupon p.a.

1.70% each quarter the basket is $\geq 100\%$; missed coupons are paid later when it recovers

100%

Capital protected

Principal returned in full at maturity whatever the basket does (Natixis credit risk only)

51%

Called at first review

Risk-neutral. 82% called within 5 years; expected life 2.11 years

We recommend **a USD 100mn, 5-year Natixis capital-protected autocallable note** on a semiconductor-heavy basket of four real, Bloomberg-listed indices (Taiwan TAIEX 35%, KOSPI 200 25%, Nikkei 225 20%, PHLX Semiconductor 20%). It gives the family the AI hardware layer (chips, memory and chip-making equipment) that complements, rather than duplicates, the infrastructure, power and property they already own. **The family never loses capital at maturity, earns 6.8% a year in the 82% of paths where the basket gets back to its starting level, and gets 100% back in the rest. Fair value is 97.7%, so issuing at par leaves Natixis 2.4% for hedging and margin.**

The family wants growth without putting the legacy at risk

Client diagnosis: third-generation dynasty, succession done, now allocating for legacy

Who they are

- Single-family office: NKE Private Wealth
- Net worth > USD 2bn; mandate USD 100mn (~5%)
- Businesses: tech infrastructure, renewable power plants, real estate in Japan and Greater China

What they want

- Capital appreciation for future generations
- Meaningful exposure to Asia's digital transformation
- To bridge traditional Asian sectors with emerging tech they know

What they can bear

- Moderate risk: preserve the legacy first
- 5-year horizon; coupons can be reinvested
- Accepts limited liquidity on 5% of wealth
- Volatile backdrop: no capital at risk

Client need	Design answer in our note
Capital appreciation	6.8% p.a. coupon, above the 5.69% 5Y USD funding rate, in every path that autocalls; proceeds roll into the next note
Exposure to Asian digital transformation	Semiconductor-heavy basket: the AI hardware layer behind the family's tech-infrastructure business (slide 3)
Generational preservation	100% capital protection at maturity, diversified basket (not worst-of), BPCE-backed issuer
Volatile markets	Memory coupon: a missed coupon is not lost but paid when the basket recovers; drawdowns cannot touch the principal

Why now: Asia builds the hardware of the AI economy

Investment thesis: structural growth themes, and a rate level that makes capital protection cheap

1 Taiwan: the AI foundry

TSMC and its supply chain make most of the world's advanced AI chips; Taiwan's index is dominated by semiconductors.

2 Korea: memory for AI

Samsung and SK Hynix supply the high-bandwidth memory every AI accelerator needs; together a large share of KOSPI 200.

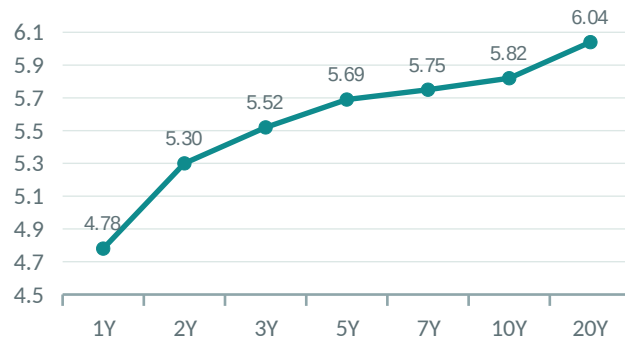
3 Japan: chip-making tools

Tokyo Electron, Advantest and peers make the equipment and testers chip factories need, alongside TSE governance reforms.

4 Global chip leaders

The PHLX Semiconductor index adds the global designers and toolmakers (incl. TSMC and ASML ADRs) on the other side of Asia's supply chain.

USD funding curve (case grid, %)

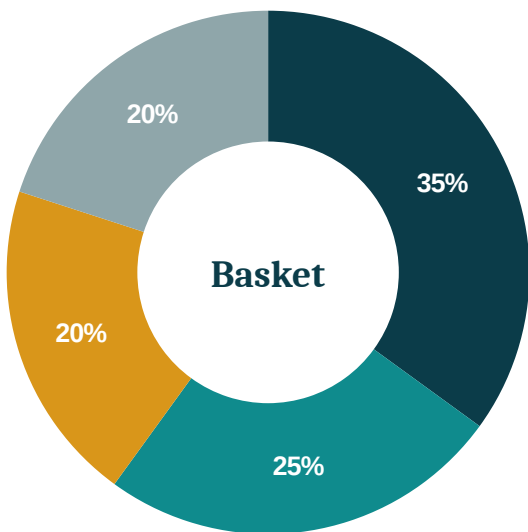


What the curve means for structuring

High USD rates make the 5Y zero-coupon bond cheap (75.8%), so guaranteeing 100% back costs little. Asian equity volatility (basket ~21%) makes it likely the basket revisits 100%, which is what pays the 6.8% coupon.

Underlying: a semiconductor-heavy basket of four indices

Bespoke weighted basket, USD quanto (FX risk sits with Natixis); all constituents searchable on Bloomberg



Index (Bloomberg)	Wt	Why it belongs (link to family)	Vol*
Taiwan TAIEX TWSE Index	35%	Foundry and chip supply chain (TSMC, MediaTek, ASE): the core of AI chip production, in the family's Greater China home region	24%
KOSPI 200 KOSPI2 Index	25%	Memory for AI (Samsung, SK Hynix): high-bandwidth memory inside every AI accelerator	25%
Nikkei 225 NKY Index	20%	Chip-making equipment and test (Tokyo Electron, Advantest): home market of the family's tech-infrastructure business	22%
PHLX Semiconductor SOX Index	20%	Global semiconductor leaders, USD-listed: diversifies beyond Asia's single-country risks	35%

Why a weighted basket, not worst-of: a shock to one market (e.g. Taiwan geopolitics) is cushioned by the other three. Basket vol is ~21% vs 35% for the PHLX Semiconductor index alone, and correlation risk stays with the desk.

*Assumed 5Y implied vols for pricing. The draft's thematic indices (e.g. 'Greater China PropTech & NLP') are not Bloomberg-listed, so we use real indices.

Term sheet: 5-year capital-protected autocallable, 6.8% coupon

Why this design: above the funding rate, with zero chance of losing capital at maturity

Issuer	Natixis SA (BPCE Group)
Notional / currency	USD 100,000,000 · quanto USD
Trade / maturity	17 Sep 2026 / 17 Sep 2031 (5Y)
Underlying	Weighted basket of 4 indices (slide 3)
Observation	Quarterly, 20 dates; autocall from Q4 (Year 1)
Coupon	6.8% p.a. (1.70% per quarter), paid if basket $\geq$ 100%
Memory	Missed coupons paid on the next date basket $\geq$ 100%
Autocall	Basket $\geq$ 100% from Q4: 100% + coupons due, note ends
At maturity	100% of principal, whatever the basket level
Issue price / fair value	100% / 97.7%

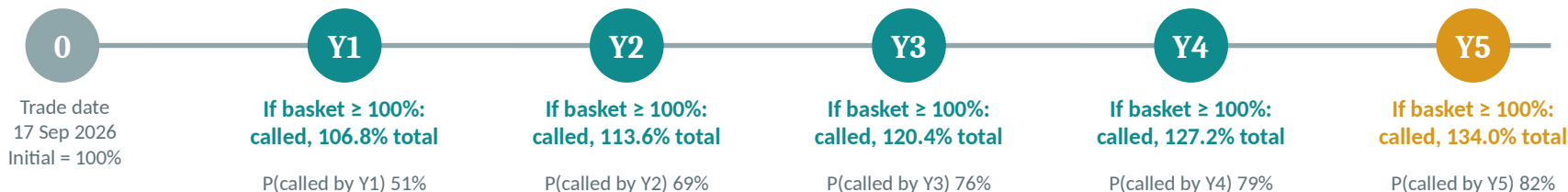
Autocall design (same basket, issued at 98%)	Fair cpn	Loss risk
Classic: fixed coupon, 70% barrier	6.3%	11.6%
Phoenix memory (80% cpn barrier), 70% barrier	7.9%	11.6%
Snowball, 65% barrier	10.8%	9.8%
Classic, 70% barrier, 10%-vol basket	4.1%	1.0%
Capital protected, memory coupon (ours)	7.1%	0.0%

Design call: The draft's 8.5% fixed coupon is worth 102.2% to the client, so Natixis cannot fund it. At-risk designs pay 6%–11% for a 10%–12% chance of losing capital. Cutting that to ~1% needs a 10%-vol basket and pays only 4.1%, below the 5.69% bond. Full protection prices at 7.1%; we offer 6.8%.

How it pays: one question every quarter

Is the basket at or above 100% of its starting level? (checked quarterly; one date per year shown)

Q1–Q3: coupon only (1.70% if basket $\geq$ 100%). From Q4: coupon plus autocall. Missed coupons are always caught up.



A Autocalled

Basket $\geq$ 100% on any date from Q4: 100% plus every coupon to date, including missed ones. The note ends.

82% risk-neutral probability

B Not called, early coupons

Basket $\geq$ 100% in Q1–Q3 only, never afterwards. The family keeps those coupons and gets 100% back at Y5.

7% risk-neutral probability

C Not called, no coupon

Basket never back to 100%. The family gets 100% back at Y5: a 0% return, but no capital loss.

11% risk-neutral probability

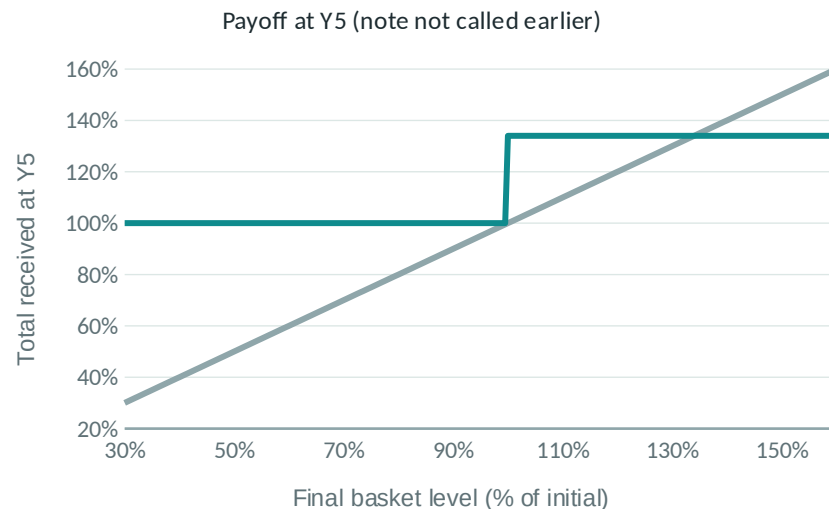
Payoff diagrams: what the family receives, and when

Left: total cash if called at each date. Right: payoff at Y5 if never called earlier, vs. owning the basket directly



Y1: 106.8% (USD 106.8mn)
Y2: 113.6% (USD 113.6mn)
Y3: 120.4% (USD 120.4mn)
Y4: 127.2% (USD 127.2mn)
Y5: 134.0% (USD 134.0mn)

With memory, the total depends only on when the note is called, not on how many coupon dates were missed along the way.



Autocallable note Basket directly

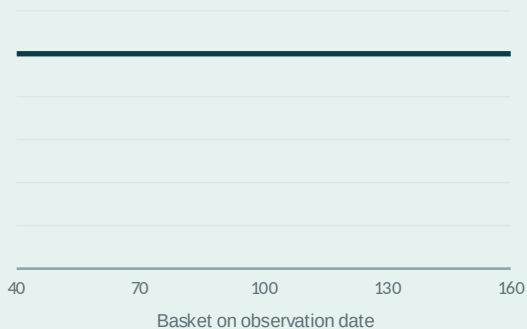
- < 100%: 100% back; capital protected however far the basket falls
- ≥ 100%: 134.0% (principal + all 20 coupons via memory)

Under the hood: a bond plus a strip of digital options

Replication used by the Natixis desk to price and hedge (values in % of notional, at trade date)

Protected principal

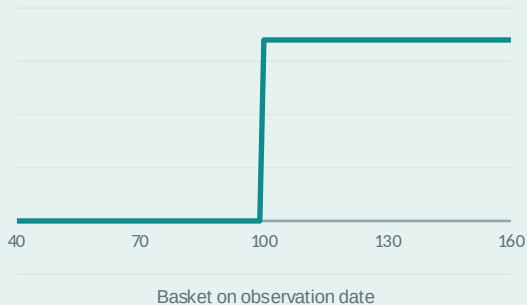
100% repaid at autocall or maturity. Worth 89.9%: early repayment is worth more than a 5Y bond (75.8%)



+

Long quarterly digitals

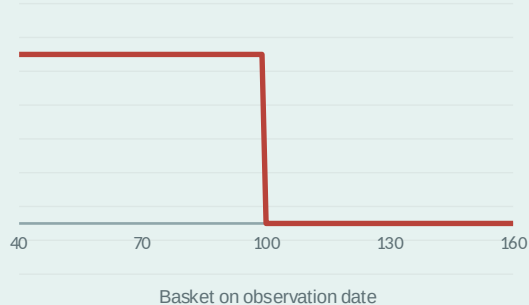
Pay 1.70%, plus missed coupons, on each date the basket is $\geq 100\%$. Worth 7.7%



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Autocall (Natixis's side)

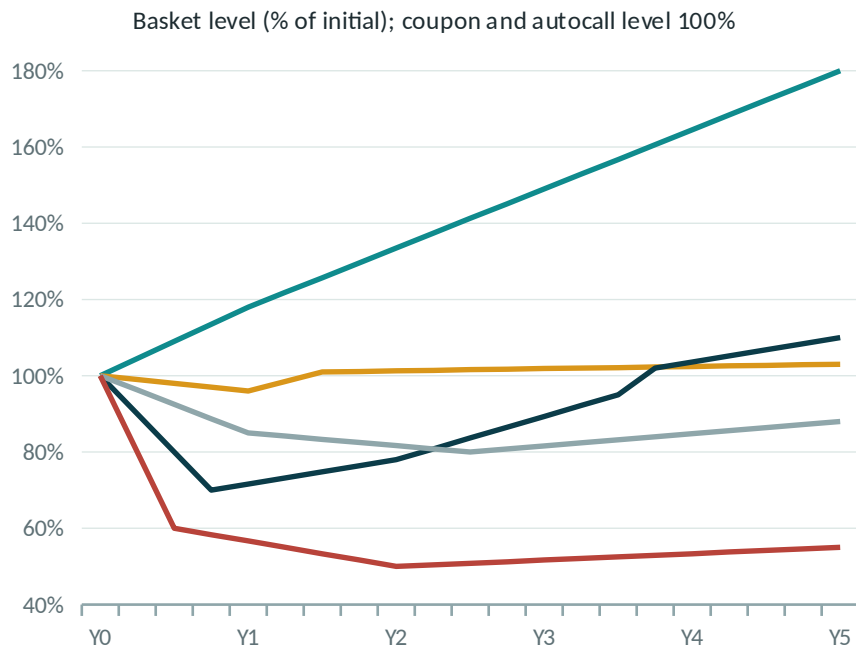
Remaining coupons: cancelled on a date from Q4 with basket $\geq 100\%$, when the note ends. This is what keeps the coupon affordable



Fair value 97.7% = principal 89.9% + coupons 7.7% (net of autocall). Issued at 100%, leaving 2.4% for hedging costs and Natixis margin. The client sells no put, so it takes no equity downside: it gives up upside above the coupon instead.

Scenario analysis: five paths for Asian markets

Hypothetical quarterly basket paths and what the USD 100mn note returns in each

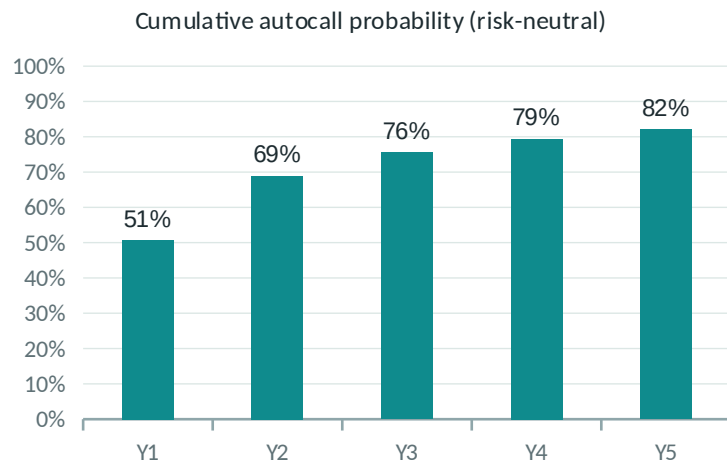


Scenario	Outcome	Coupons	USD mn	IRR	Basket
AI capex boom	Called Q4 (1y)	6.8%	106.8	7.0%	–
Range-bound Asia	Called Q6 (1.5y)	10.2%	110.2	6.7%	–
2022-style tech drawdown, slow recovery	Called Q15 (3.75y)	25.5%	125.5	6.2%	–
Lost decade (sideways below par)	100% back at Y5	0.0%	100.0	0.0%	-12%
Geopolitical shock, no recovery	100% back at Y5	0.0%	100.0	0.0%	-45%

Reading: the note earns ~6.2%–7.0% a year whenever the basket gets back to 100%, even after a 30% drawdown: memory pays all 15 missed coupons at Q15. In a lost decade or a lasting shock the family gets 100% back, while the basket itself is down 12–45%.

Simulation back-test: most paths end early with full coupons

Monte Carlo outcome distribution: risk-neutral and under three real-world equity return assumptions



Metric	Risk-neutral	0% equity p.a.	4% equity p.a.	8% equity p.a.
Called within 5Y	82%	69%	80%	88%
Expected life (yrs)	2.1	2.6	2.2	1.9
Mean IRR	5.6%	4.8%	5.5%	6.0%
Median IRR	6.9%	6.8%	6.9%	6.9%
Avg coupons received	8.4%	7.3%	8.2%	8.6%
Zero return (100% back)	11%	20%	13%	8%
Capital loss	0%	0%	0%	0%

51%

called at the first review
(106.8% total)

0%

chance of losing capital at
maturity

11%

of paths earn nothing:
100% back after 5 years

Risks for the family, and how Natixis hedges its side

Transparent disclosure plus a desk hedging plan for every exposure

Risk	What could happen	Mitigant
Opportunity cost	Basket never back to 100%: 0% for 5 years (~11% of paths)	Coupon above the funding rate in most paths; only 5% of wealth
Capped upside	Basket +80% still pays 6.8% p.a. and is called at Y1	Rest of family wealth keeps direct equity exposure
Reinvestment	51% called at Y1: cash back at prevailing rates	Natixis offers a roll into a new note at call
Issuer credit	Protection depends on Natixis / BPCE paying	BPCE senior rating; optional collateralised wrapper
Liquidity / MTM	Sold before maturity, the price can be below 100%	Natixis daily indicative price; hold to call or maturity

Pricing sensitivity (fair coupon): base 7.1% · vol ±3pts 6.8%–7.4% · corr ±0.15 6.9%–7.3% · equity drift –1/–2% 7.7%–8.2%. The 6.8% offer stays fundable in every case.

Coupon digitals

Natixis owes basket digitals struck at 100%: replicated with tight basket call spreads, delta-hedged with TAIEX / KOSPI 200 / Nikkei / SOX futures

Vol & correlation

Coupon value moves with basket vol and correlation (fair 6.8%–7.4% for vol ±3pts): managed with listed index options and correlation trades

Quanto FX

USD payout on TWD/KRW/JPY indices (SOX already USD): FX forwards and quanto adjustment in price

Rates & funding

Protected principal funded at the 5.69% curve and hedged with USD swaps; the autocall shortens the liability

Appendix A: pricing assumptions and methodology

Inputs to refresh on Bloomberg at trade date (BVOL / OVDV for vols, CORR for correlations)

Index	Weight	Vol	Div	TWSE	KOSPI2	NKY	SOX
TWSE	35%	24%	2.8%	1.00	0.70	0.60	0.55
KOSPI2	25%	25%	2.0%	0.70	1.00	0.55	0.45
NKY	20%	22%	1.8%	0.60	0.55	1.00	0.40
SOX	20%	35%	0.8%	0.55	0.45	0.40	1.00

Model. Correlated GBM for each index, 200,000 paths, quarterly steps. Risk-neutral drift = USD funding (linear interpolation of case grid) – dividend yield. Quanto adjustment ignored (assumed hedged by the desk; to be added with FX-equity correlations).

Discounting. USD funding grid from the case rules; 5Y discount factor 0.758.

Coupon solve. Bisection on coupon so that PV = 98% (2% hedging cost and margin).

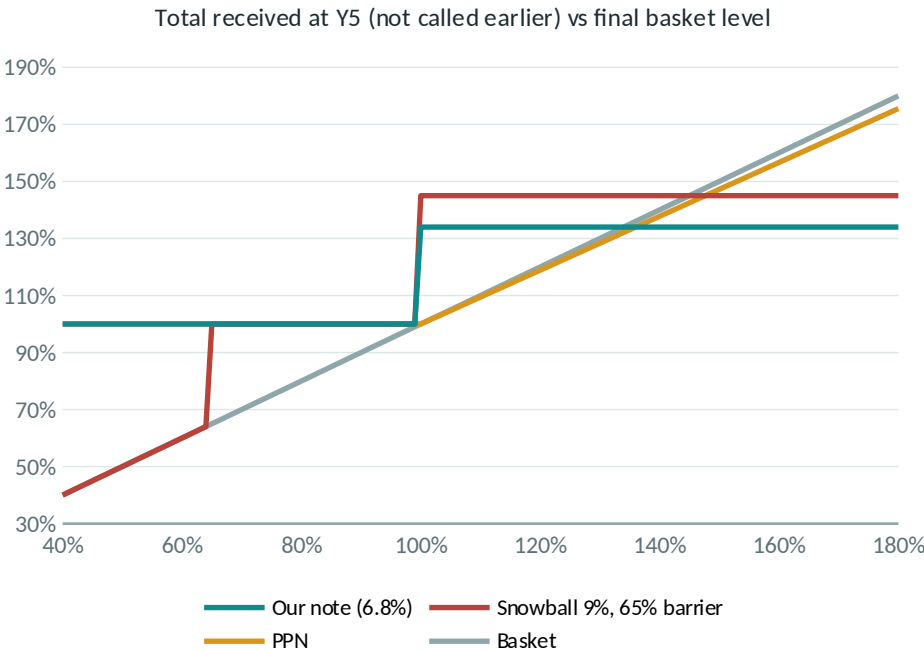
Historical back-test (to run). Monthly Bloomberg history since 2015 for all four indices. Launch a hypothetical note every month, apply the same coupon, memory and autocall rules, and record coupons, life and IRR by launch year.

Design variant (capital protected)	Fair cpn
Coupon barrier 100% (base)	7.1%
Coupon barrier 95%	6.5%
Coupon barrier 90%	6.0%
First autocall at Y2	7.3%
Equity drift –1%	7.7%
Equity drift –2%	8.2%
Vol +3pts / –3pts	7.4% / 6.8%

All variants priced at 98% with the same basket and inputs. Code: pricing/scenario2_autocall_mc.py

Appendix B: alternatives for the investment committee

Same basket: a higher-coupon at-risk snowball, and a pure participation note



Option	Return	Loss risk
Protected autocallable (ours)	6.8% p.a. memory coupon	0%
Snowball, 65% barrier	9% p.a., paid at call	9.8%
Principal-protected note	94% of basket gain at Y5	0%

Trade-off: The snowball pays 9% but loses capital in ~10% of paths. The PPN keeps full protection with uncapped upside, but pays nothing unless the basket ends above 100% after 5 years and has no early exit. Our note sits between them: full protection, a coupon above the funding rate, and money usually back within ~2 years.